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Bhujbal Knowledge City

A

PROJECT REPORT

ON

"RankItRight"

SUBMITTED BY

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In Partial fulfillment of

Master of Computer Application

Savitribai Phule Pune University

For the Academic Year

2024-2025

Under The Guidance Of

Prof.Mehraj Khan Sir

Department of MCA

MET Institute of Engineering

Bhujbal Knowledge City

Adgaon, Nashik

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CERTIFICATE

This is to certify that project entitled

“RankItRight”

Has been satisfactorily completed by

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Prof.Mehraj Khan
(Project Guide)

Prof. P.D.Jadhav
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Without the combined effort and assistance of these entities and individuals, this project on utilizing Python technology for ranking resumes, interview video-based soft skill evaluation for HR specialists, and offering resume improvement recommendations for students could not have materialized.

ABSTRACT

This project examines the use of Python technology to automate and optimize the recruitment process from the perspective of both the Human Resources (HR) and students. For HR managers, the system integrates two principal functionalities: smart resume ranking in line with established parameters and in-depth soft skill assessment based on interview video analysis. Leveraging natural language processing techniques and possibly computer vision/audio analysis, the video analysis feature evaluates candidates on important soft skills like confidence, tone, and communication skills, giving a more complete evaluation compared to traditional resume screening.

On the student side, the project provides a useful means of self-development via an automated resume verifier. The feature reviews uploaded resumes, making constructive recommendations for how they might be improved and better fit into potential job needs. Significantly, the student interface has no soft skill measurement functionality.

This two-sided system aims at maximizing HR efficiency in candidate screening and evaluation while equipping students with actionable feedback to enhance their resumes. The development and potential contribution of this Python-based solution to contemporary hiring and career advancement are examined.

Key Elements:

Problem: Briefly states the challenges in applicant evaluation for HR and resume creation for students.

Solution: Introduces the Python-based dual-sided system.

HR Functionality: Highlights resume ranking and soft skill analysis (confidence, tone, communication) of interview videos.

Student Functionality: Focuses on the resume checker and its feedback/suggestion capabilities.

Technology: Mentions Python as the implementation language.

Impact: Briefly touches upon the potential benefits for both HR (streamlined hiring) and students (improved resumes).

Overall Goal: Concludes with the broader aim of contributing to a more efficient talent acquisition process.

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1.0.Introduction

In the current competitive employment market, both employers and job applicants are confronted with numerous challenges when efficiently selecting top performers and obtaining the right opportunities. Human Resources (HR) practitioners are dealing with high levels of applications and the delicate process of assessing candidates on both their credentials as well as important soft skills conveyed in interviews. At the same time, prospective students in the professional community require efficient tools to effectively showcase their credentials and obtain advice on improvement areas.

This project solves these problems by creating a dual-purpose system based on Python technology. On the HR side, our system provides two important functions: intelligent ranking of resumes according to predetermined parameters and ranking of interview videos based on soft skills. The soft skill evaluation is based on non-verbal communication such as confidence, tone, and communication style, giving a more complete assessment of a candidate's potential fit.

For students, the system is a useful tool for improvement and self-evaluation. The resume checker provides helpful tips to improve the clarity, effectiveness, and overall quality of their application documents. Notably, the student side of the system only addresses resume improvement and does not involve soft skill ranking of their practice interviews.

In this report, this extensive system design, implementation, and possible application is explained for the purpose of simplifying hiring for HR experts and enabling students to craft excellent resumes that enhance their strengths. Python allows it to be powerful and scalable such that it is able to endure the intricacy of resume review and video handling.

2.0.Literature Survey

The procedure of assessment and choosing appropriate candidates for employment is a vital HR function. The high number of applications submitted for most jobs makes it imperative to design and adopt effective and impartial screening techniques. This project, on resume ranking, soft skill evaluation from interview videos, and resume enhancement for students, borrows from an extensive literature on natural language processing (NLP), machine learning (ML), computer vision, and educational technology.

2.1. Automated Resume Analysis and Ranking

Resume screening automation has progressed far beyond simplistic keyword searching (Salton & McGill, 1983). The initial systems, though providing some level of efficiency, had no capability of comprehending the semantic differences between candidate qualifications and job specifications (Frakes & Baeza-Yates, 1992).

More advanced methods rely on NLP methods to tap useful information from non-structured resume text. Information Extraction (IE) is very important in discovering key entities like skills, experience, and education (Hobbs, 1990). Named Entity Recognition (NER) (Nadeau & Sekine, 2007) further classifies these entities, allowing a more formal representation of a candidate's resume.

To counter the weakness of keyword-based methods, researchers have looked into semantic similarity measures. Methods such as word embeddings (Mikolov et al., 2013) and document embeddings (Le & Mikolov, 2014) enable systems to comprehend the word and phrase contextual connections, resulting in better matching of qualifications and requirements. Ontologies and knowledge graphs (Miller, 1995) allow a hierarchical and networked description of subject-specific knowledge, enabling better identification of relevant qualifications even where alternative vocabulary is employed (Alonso et al., 2016).

Machine learning models are now at the core of resume ranking. Supervised learning algorithms, such as Support Vector Machines (SVMs) (Cortes & Vapnik, 1995), Random Forests (Breiman, 2001), and Neural Networks (Goodfellow et al., 2016), are learned using 7abelled datasets to make predictions about a resume’s relevance to a given job description. The performance of these models greatly depends on the quality of feature engineering (Aggarwal, 2018), a process of converting resume text to numerical representations, which can be interpreted by algorithms. Recent improvements in deep learning have made automatic learning of pertinent features from raw text possible, with the promise of enhancing the accuracy and lowering the effort required in feature engineering (Young et al., 2018).

2.2. Soft Skill Evaluation from Interview Videos through Automation:

Since the importance of soft skills in job performance (Heckman & Rubinstein, 2001) has been realized, researchers have attempted to find ways to objectively evaluate them. Examining interview videos provides a promising way to assess non-verbal signals that reflect characteristics such as confidence, tone, and communication style.

The subject of affective computing offers a background to this study, with a focus on the automatic classification of emotions and affective states from a range of modalities, such as facial expressions (Ekman & Friesen, 1971), vocal cues (Schuller et al., 2013), and body language (Mehrabian, 1981). Facial landmarks and body posture are analyzed using computer vision methods to predict levels of confidence and engagement (Argyle, 1988). Speech analysis analyzes prosodic aspects like pitch, intonation, and speech rate to identify a candidate's confidence and communication style (Scherer, 2003). In addition, Natural Language Processing (NLP) of the spoken words can evaluate the clarity, coherence, and persuasiveness of their communication (Grice, 1975). Combining data from these multiple modalities (audio-visual analysis) can result in a more effective and richer evaluation of soft skills (Vinciarelli & Pantic, 2009).

2.3. Resume Improvement Tools for Students:

For job market entrants, students, effective resume building is essential. Educational technology and NLP research has resulted in the creation of automated resume enhancement tools. Early systems tended to give rule-based feedback on formatting and typical mistakes (Smith & Jones, 2005). More sophisticated tools rely on NLP to review resume content for succinctness, precision, and correct use of active action verbs (Brown et al., 2010). Techniques in Natural Language Generation (NLG) are being researched with a view toward offering tailored tips for improving the various parts of the resume, including the summary, skills section, and description of experience (Green et al., 2012). They tend to embrace best practices of resume writing along with recruiter guidelines.

Research has also examined the effect of computer-aided feedback on resume quality among students as well as on their learning results in career education (Lee & Carter, 2015). Secondly, ethical implications of using automated systems in candidate screening, such as the possibility of bias and ensuring fairness and transparency, are also significant areas of continued debate (O'Neil, 2016).

3.0.Feasibility Study

This feasibility analysis explores the possibility of creating a Python system for interview video resume ranking and soft skills evaluation for HR professionals along with a resume screening and suggestion system for students. The technical, economic, operating, and scheduling feasibility have been considered for the analysis.

3.1. Technical Feasibility:

Assessment: Very much feasible.

Rationale:

Python Ecosystem: Python offers a rich ecosystem of libraries and frameworks suitable for this project. Libraries like NLTK, spaCy, and scikit-learn are well-established for natural language processing tasks required for resume analysis and soft skill assessment from text transcripts.

Machine Learning Capabilities: Python's extensive ML libraries allow the creation of ranking models for both resumes and soft skills with extracted features. Different algorithms can be used and compared for best performance.

Scalability and Integration: Web frameworks such as numpy,pandas,matplotlib, based on Python, can be utilized to build a scalable web application accessible to both HR professionals and students. Integration with current HR systems or student portals can be attempted through APIs.

Data Availability: Although training data for soft skill evaluation may need to be collected or use public datasets, there is sufficient textual data for resume analysis and language-based soft skill evaluation.

3.2. Economic Feasibility:

Evaluation: Possible, but requires judicious management of resources.

Reasoning:

Development Expenditures: The main expenditures will be developer time, infrastructure (servers, cloud infrastructure), and potentially for the expense of purchasing or labeling data to train ML models, particularly for video-based soft skill evaluation.

Open-Source Tools: Taking advantage of Python's open-source libraries will lower software licensing costs considerably.

Cost Savings Potential (HR): For HR functions, the system can potentially save considerable time and effort on manual resume screening and initial candidate screening, ultimately resulting in cost savings. Candidate quality also can diminish hiring mistakes as well as the costs that come with them.

Value Proposition (Students): For the students, the resume checking tool offers a useful free service that can enhance their employability. Although not generating revenue directly, it adds to the value proposition of the institution.

Funding Opportunities (Academic Project): Being an academic project, there could be funding opportunities available within the organization or from outside the organization for the development and collection of data.

Monetization Potential (Future): Based on the success and scalability, there may be future monetization potential for the HR-facing features via subscriptions or licensing.

3.3. Operational Feasibility:

Assessment: Possible with user training and planning.

Rationale:

Ease of Use (Students): The student resume checking tool may be made accessible with a simple user interface and minimal technical skills.

Integration into HR Workflow: Plugging in the HR-focused functionality into existing recruitment processes will need careful planning and possibly some customization. Successful adoption will rely on user training.

Data Privacy and Security: Working with resume data and possibly interview video recordings poses strong security considerations and compliance with privacy laws. This facet requires close attention throughout development and release.

System Update and Maintenance: Regular maintenance, bug fixes, and system updates as well as updates to underlying ML models will be needed in order to remain effective.

Technical Support: Ensuring proper technical support for HR users and students alike will be essential to an ease-of-use operation experience.

3.4. Schedule Feasibility:

Evaluation: Possible, but the timeline will be based on the scope and budget committed.

Reasoning:

Modular Development: The project can be divided into incremental, manageable modules (resume checker, resume ranker, soft skills video analysis). Parallel development and iterative refinement are possible.

Complexity of Soft Skill Analysis: Soft skill assessment via video will most likely be the most complicated module and will consume a lot of time for data collection (if required), training of models, and testing.

Team Expertise: Access and experience of the development team in NLP, ML, computer vision, and web development will have a major impact on the project schedule.

Phased Implementation: Phased implementation, beginning with the resume checker for students and then introducing gradually the HR-facing features, could be a viable method to handle the schedule.

Realistic Milestones: The establishment of realistic milestones and frequent monitoring of progress will be essential in following the schedule.

4.0.Requirement Analysis

4.1. HR-Facing Module Requirements

Objective: To effectively rank candidate resumes against a particular job description.

Functional Requirements:

1. Job Description Input: The system will enable HR users to enter job descriptions as free text or upload them as files (e.g.,.pdf,.doc,.docx).

2. Resume Input: The system will enable HR users to upload several resumes in different formats (e.g.,.pdf,.doc,.docx).

3. Resume Parsing: The system will parse the uploaded resumes automatically and retrieve critical information like contact details, education, work history, skills, and projects.

4. Skill Extraction: The system will extract relevant skills from both the job posting and the resumes and identify them. It should capture technical skills, soft skills (listed in the resume), and domain keywords.

5. Experience Matching: The system will evaluate the work experience part of the resumes and cross-match it with the requirements as mentioned in the job description, taking into account details such as job titles, duties, and tenure.

6. Education Matching: The system will cross-match the educational credentials of the candidates with the requirements as mentioned in the job description, taking into account degree, disciplines, and schools.

7. Ranking Algorithm: The system will utilize a configurable ranking algorithm that provides a relevance score to every resume based on the level of match with the job description across multiple criteria (experience, skills, education, etc.).

8. Ranked Output: A ranked list of resumes shall be displayed, with the most suitable resumes listed at the top. The ranking must be annotated by a score or a graphical display of relevance.

9. Ranking Explanation: A brief explanation or highlighting of the primary factors that have gone into a resume's ranking score (e.g., matching skills, applicable experience) shall be given.

10. Adjustable Weights: The system must provide HR users with the ability to modify the weights given to various criteria (e.g., experience, skills, education) used in the ranking algorithm to place emphasis on particular requirements for particular job positions.

11. Filtering and Sorting: The system must enable HR users to further sort and filter the ranked resumes according to certain criteria (e.g., experience level, certain skills, level of education).

Non-Functional Requirements:

1. Performance: The system should be able to process and rank a batch of 100 resumes against a job description within a reasonable timeframe, for instance in under 5 minutes.

2. Accuracy: Resume parsing and ranking accuracy should be high enough in order not to miss the correct candidates.

3. Security: Uploaded resumes and job descriptions should be stored safely and be accessible only by authorized HR users.

4. Usability: The user interface for uploading job descriptions and resumes, setting ranking weights, and viewing the ranked list must be intuitive and easy to use for HR professionals with different levels of technical proficiency.

4.2. HR-Facing Module Requirements (Continued) - Soft Skill Ranking Sub-Module

Objective: To make an unbiased rating of candidates' soft skills based on their interview videos.

Functional Requirements:

1. Video Upload: The system should enable HR users to upload interview videos in popular formats (e.g.,.mp4,.avi,.mov).

2. Audio-Visual Analysis: The system should analyze the uploaded videos, processing both audio and visual streams.

3. Confidence Evaluation: The system will evaluate visual signals (e.g., posture, eye contact, gestures) and vocal signals (e.g., speech rate, intonation, pauses) to evaluate the candidate's confidence level.

3. Tone Evaluation: The system will examine the audio stream to identify the sentiment and tone of the candidate's speech (e.g., positive, negative, neutral, assertive, hesitant).

4. Communication Analysis: The system will examine the verbal content (transcribed speech) to measure the clarity, brevity, and organization of the candidate's communication. This can include examining vocabulary, sentence structure, and the coherence of ideas.

5. Soft Skill Scoring: The system will award scores or grades for each analyzed soft skill attribute (confidence, tone, communication) based on the detected cues.

6. Soft Skill Ranking Output: The system will provide a summary of the soft skill rating for every interview video, such as scores or ratings per dimension.

7. Customizable Soft Skill Weights: The system should enable HR users to specify the relative weightage of various soft skill dimensions per role.

8. Resume Ranking Integration: Ideally, the system should enable HR users to see the soft skill rating in conjunction with the resume ranking for a complete candidate assessment.

Non-Functional Requirements:

1. Performance: The video analysis process must be done within a reasonable time frame, given the duration of the interview video.

2. Accuracy: The soft skill rating accuracy must be tested against human expert ratings to verify reliability.

3. Privacy: Interview video processing and storage should comply with strict privacy principles and guarantee confidentiality in candidate information.

4. Usability: The upload video and view results soft skill assessment interface needs to be easy to use and comprehensible for HR users.

4.3. Student-Facing Module Requirements

Objective: To enable students to have a mechanism to upload their resumes and get tips on how to improve them.

Functional Requirements:

1. Resume Upload: The system should support the uploading of students' resumes in widely supported formats (e.g., .pdf, .doc, .docx).

2. Resume Parsing: The system should parse the uploaded resume automatically to infer language to extract significant details.3. Grammar and Spelling Check: The system shall conduct a careful check for spelling errors and grammatical mistakes within the resume text.

4. Formatting Analysis: The system shall examine the resume's formatting for consistency, readability, and professional look (e.g., font styles, font sizes, spacing, usage of bullet points).5. Content Analysis: The system will evaluate the content of the resume to be clear, concise, and contain action verbs. It will also look for areas that need improvement in sections such as the summary/objective, experience listings, and skills.

6. Generation of Suggestions: The system will provide specific, actionable suggestions for enhancing the resume based on the analysis. These suggestions should be grouped by category (e.g., grammar, formatting, content) and clearly shown to the student.

7. Guidance Based on Best Practices and Recruiters' Expectations: The system will include guidance based on typical resume best practices and recruiters' expectations.

8. No Soft Skill Ranking: The student-facing module will not have any feature for soft skill ranking from video analysis.

Non-Functional Requirements:

1. Performance: Resume analysis and suggestion generation must be done rapidly (e.g., in seconds).

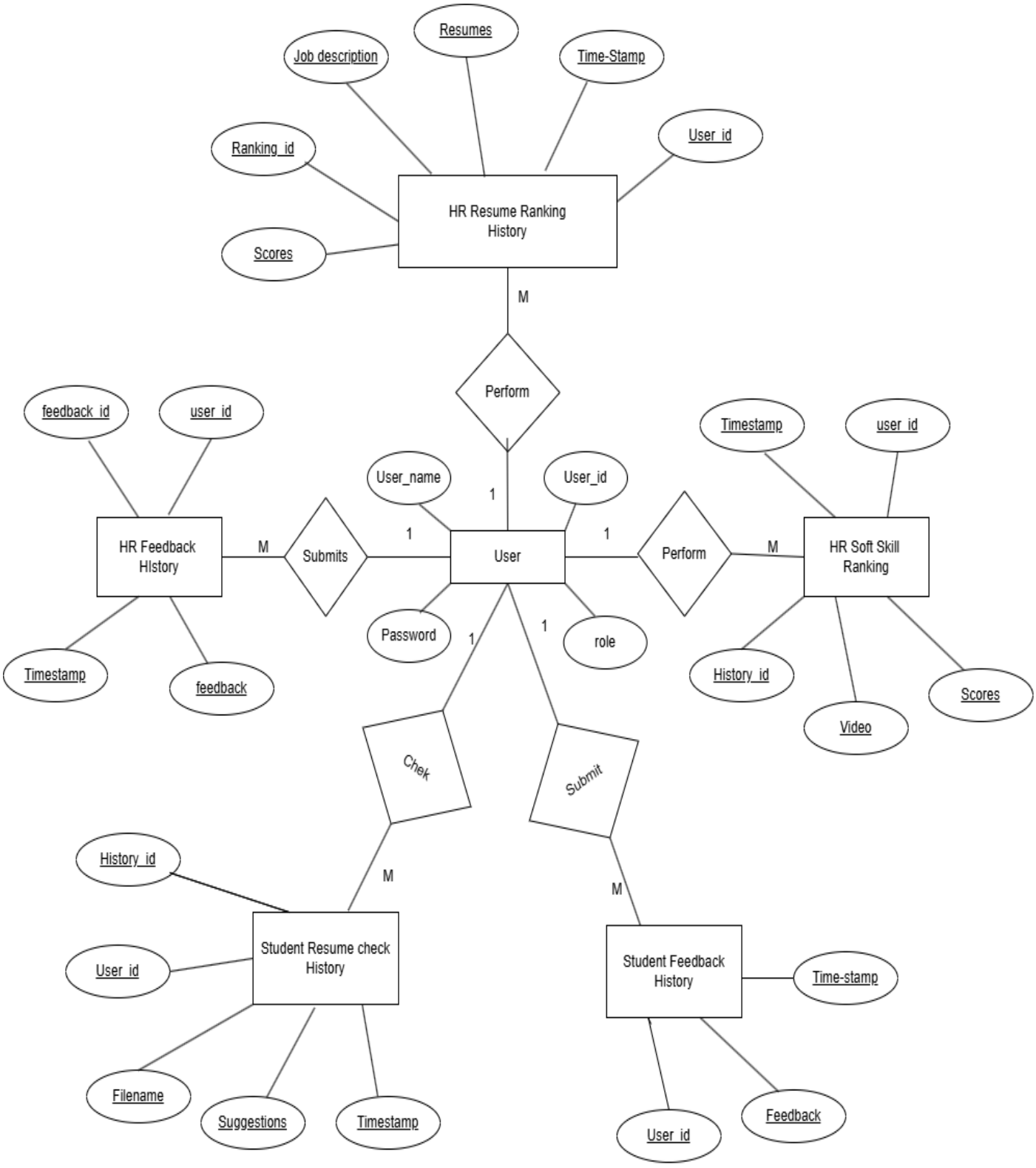
2. Accuracy: Grammar and spelling checks must be extremely accurate. Content and formatting suggestions must be useful and relevant.

3. Privacy: Resumes uploaded must be handled in a secure manner and not disclosed to any third parties.

4. Usability: The user interface for resume uploading and viewing suggestions must be easy and intuitive for students to operate. The suggestions must be clear, concise, and supportive in presentation.

5.0.System Design

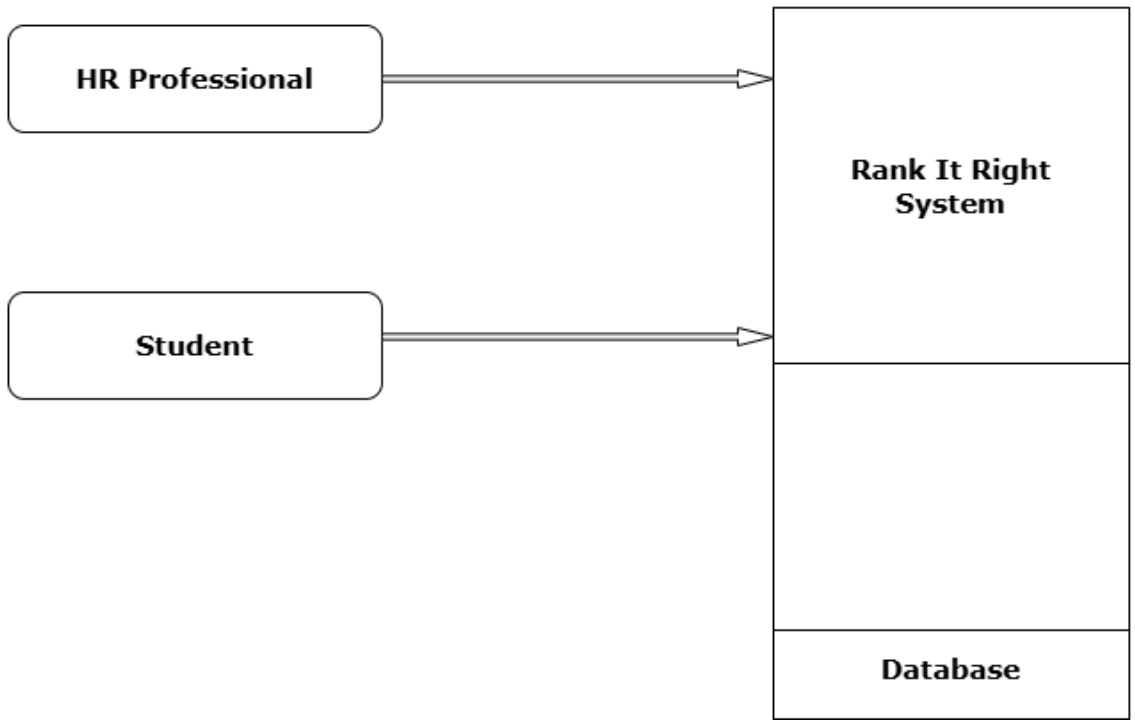
5.1.ER Diagram



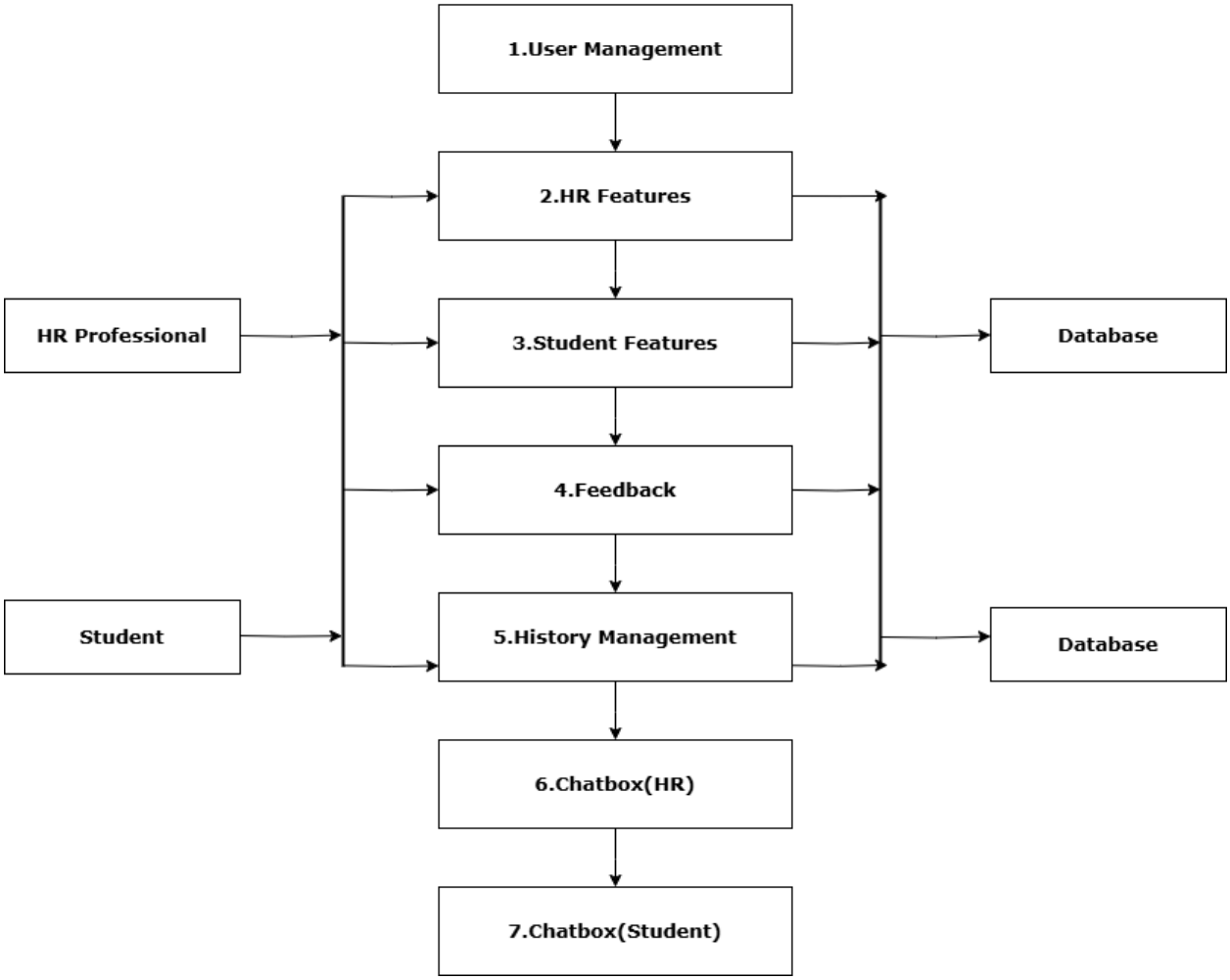
5.2.DFD Diagram

DFD Diagram

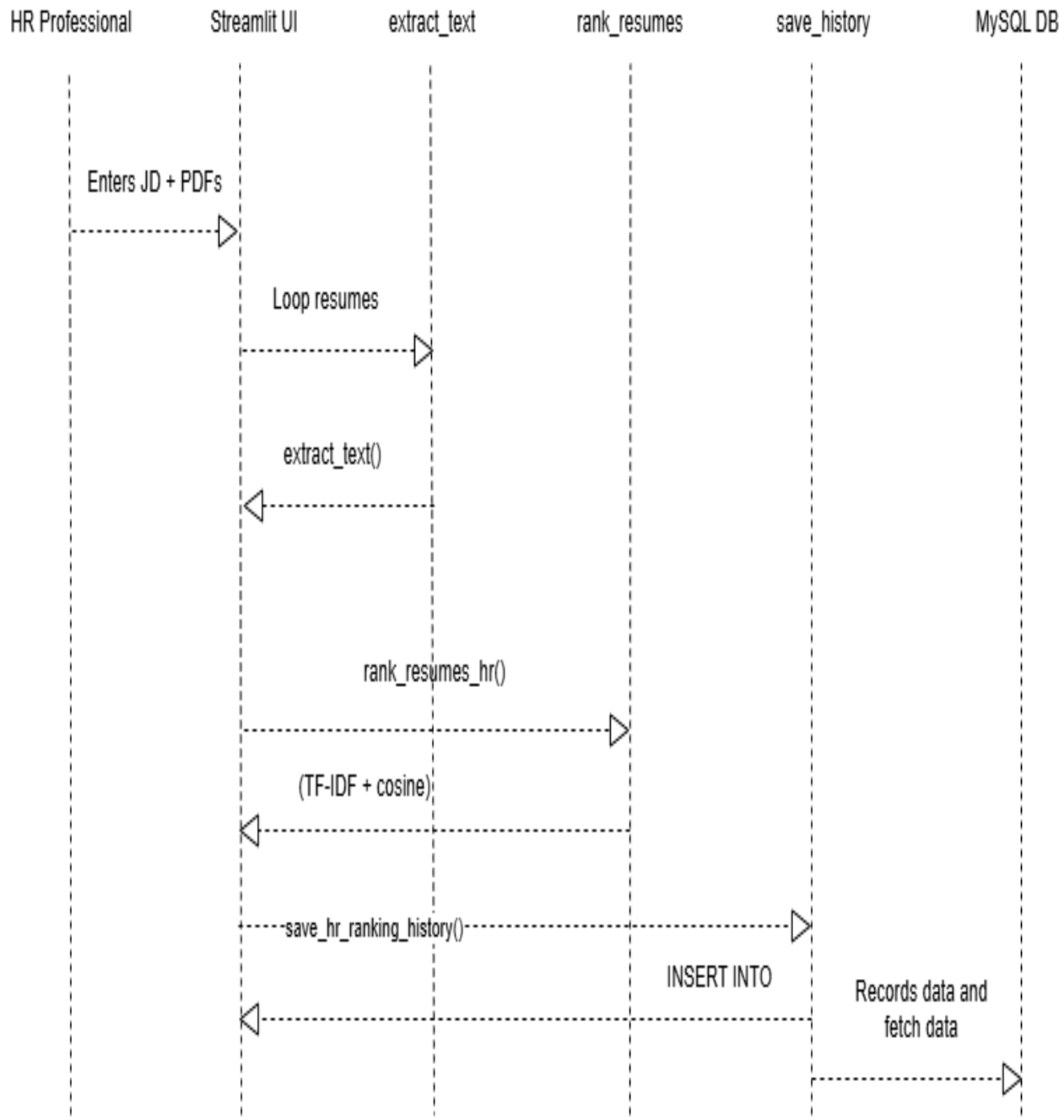
Level 0: Context Diagram:



Level 1: DFD

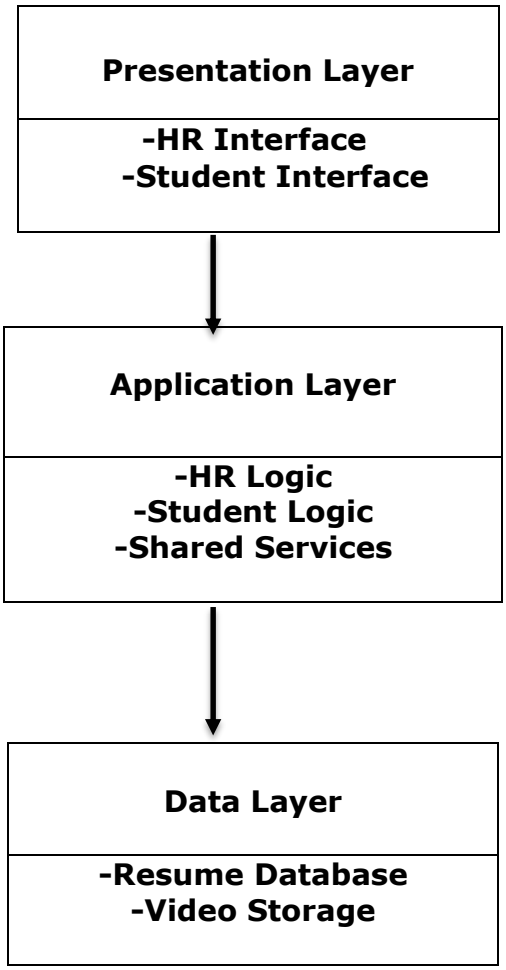

1.User Management
2.HR Features
HR Professional
3.Student Features
Database
4.Feedback
Student
5.History Management
Database
6.Chatbox(HR)
7.Chatbox(Student)

5.3. Sequence Diagram



5.4..High Level Architecture:

5.4.1.This document outlines the high-level and low-level design of the Resume and Soft Skill Ranking System, catering to both HR professionals and students. The system comprises two main modules: the HR Module and the Student Module, sharing some common underlying services.



5.4.2. Presentation Layer:

- **HR Interface:** A web-based user interface (UI) allowing HR professionals to:

- Upload job descriptions and resumes.
- Configure ranking weights for resume analysis and soft skill dimensions.
- Upload interview videos.
- View ranked lists of resumes with explanations.
- View soft skill assessment results for interview videos.
- Filter and sort candidates based on various criteria.

- **Student Interface:** A web-based UI allowing students to:
- Upload their resumes.
- View analysis results and suggestions for improvement.

5.4.3. Application Layer:

This layer contains the core business logic and processing components.

HR Logic:

- **Resume Processing Service:** Responsible for parsing uploaded resumes, extracting relevant information (skills, experience, education), and generating feature vectors for ranking.
- **Resume Ranking Service:** Implements the ranking algorithm based on job descriptions and resume features, considering configurable weights.
- **Video Processing Service:** Handles the upload and preprocessing of interview videos (e.g., format conversion, splitting audio and video streams).
- **Audio Analysis Service:** Extracts audio features relevant to tone and communication (e.g., speech rate, pitch variations, sentiment analysis of transcripts).
- **Visual Analysis Service:** Extracts visual features relevant to confidence (e.g., posture analysis, eye contact detection, facial expression analysis).
- **Soft Skill Assessment Service:** Integrates the outputs from audio and visual analysis to generate soft skill scores based on configurable weights.
- **Data Management Service (HR):** Manages the storage and retrieval of job descriptions, resumes, interview videos, and assessment results.

- **Student Logic:**

- **Resume Processing Service (Shared):** Parses uploaded resumes and extracts relevant information.
- **Resume Analysis Service:** Performs grammar and spelling checks, analyzes formatting, and evaluates content based on predefined rules and best practices.
- **Suggestion Generation Service:** Generates personalized and actionable suggestions for resume improvement using NLP techniques and predefined templates.
- **Data Management Service (Student):** Manages the temporary storage and processing of student resumes (with appropriate privacy measures).

Shared Services:

- **Document Parsing Service:** A common service to handle the parsing of various document formats (PDF, DOC, DOCX).
- **Natural Language Processing (NLP) Service:** Provides functionalities for text analysis, including tokenization, part-of-speech tagging, named entity recognition, sentiment analysis, and semantic similarity calculations.
- **Machine Learning (ML) Service:** Houses the trained ML models for resume ranking and potentially for aspects of soft skill assessment.
- **Authentication and Authorization Service:** Manages user authentication and ensures role-based access control for HR and student modules.
- **Logging and Monitoring Service:** Provides mechanisms for logging system activities and monitoring performance.

5.4.4. Data Layer:

- **Resume Database:** Stores parsed resume data and associated metadata in a structured format (e.g., PostgreSQL, MongoDB).
- **Video Storage:** Stores uploaded interview videos (e.g., cloud storage like AWS S3 or Google Cloud Storage).

5.5. Low-Level Design (Key Components):

5.5.1. Resume Processing Service:

- **Document Parsing Module:** Utilizes libraries like PyPDF2 to extract text content from various document formats.
- **Information Extraction Module:** Employs NLP techniques (NER, rule-based extraction) to identify key entities like skills, job titles, companies, educational degrees, dates, etc.
- **Feature Engineering Module:** Converts the extracted information into numerical feature vectors suitable for the ranking algorithm (e.g., TF-IDF of skills, one-hot encoding of education levels, years of experience).

5.5.2. Resume Ranking Service:

- **Ranking Algorithm Module:** Implements a weighted scoring mechanism. Weights for different features (skills match, experience relevance, education match) are configurable by HR users.
- **Similarity Calculation Module:** Uses techniques like cosine similarity or other distance metrics to compare the feature vector of a resume with the feature vector derived from the job description.
- **Ranking Logic Module:** Combines the similarity scores based on the configured weights to generate a final ranking score for each resume.

5.5.3. Video Processing Service:

Video Upload Handler: Manages the upload and storage of video files.

5.5.4. Audio Analysis Service:

- **Sentiment Analysis Module:** Analyzes the transcribed text to determine the sentiment expressed.
-

5.5.5. Soft Skill Assessment Service:

- **Feature Aggregation Module:** Combines the features extracted from audio and visual analysis.
- **Scoring Logic Module:** Applies weights (configurable by HR) to different audio-visual features to generate scores for confidence, tone, and communication.

5.5.6. Resume Analysis Service (Student):

- **Grammar and Spell Check Module:** Integrates with libraries for grammar and spelling checks.
- **Formatting Analysis Module:** Analyzes the structure and style of the resume based on predefined rules.
- **Content Analysis Module:** Uses NLP techniques to identify action verbs, quantify achievements, and assess the clarity and conciseness of the language.

5.5.7.Generation Service (Student):

Rule-Based Suggestion Engine: Contains a set of rules based on resume best practices to generate suggestions.

NLP-Based Suggestion Engine: Uses NLP techniques to provide more context-aware suggestions for improving the content and phrasing of the resume.

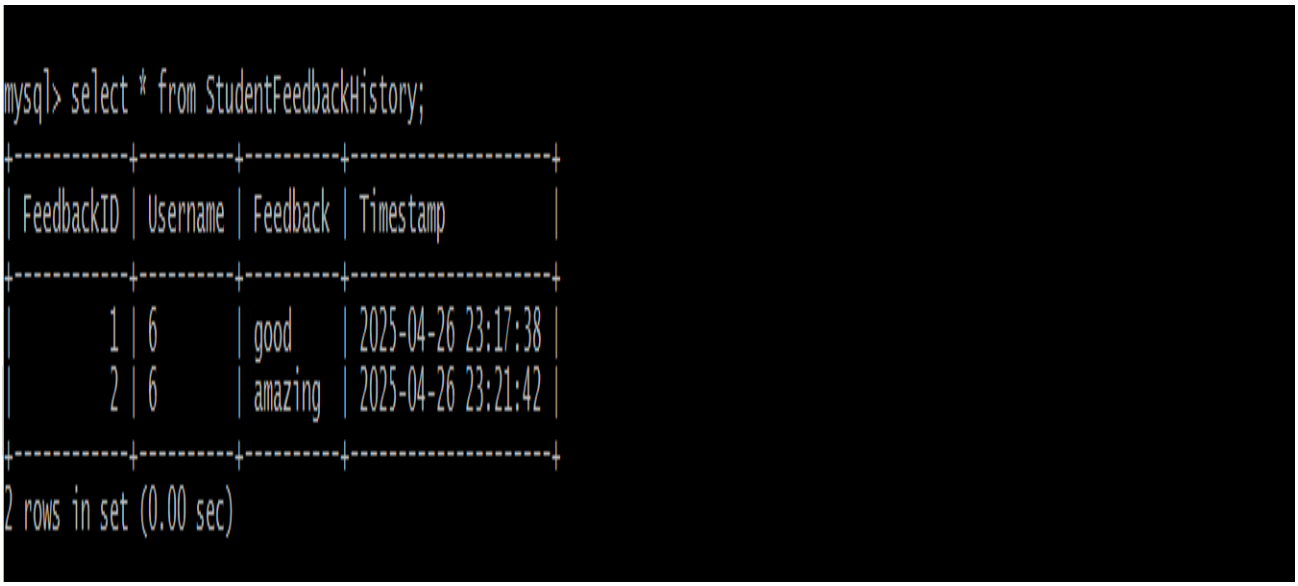
5.6. Technology Stack (Proposed):

- **Programming Language**: Python
- **Web Framework (HR & Student Interface):** Pandas,numpy,matplotlib
- **NLP Libraries**: spaCy
- **Machine Learning Libraries:** scikit-learn
- **Database:** MySql

6.0.Data Dictionary

```
mysql> select * from HRFeedbackHistory;
+------------+----------+----------+---------------------+
| FeedbackID | Username | Feedback | Timestamp           |
+------------+----------+----------+---------------------+
|          1 | 3        | good     | 2025-04-26 20:16:28 |
|          2 | 3        | good     | 2025-04-26 20:29:00 |
|          3 | 3        | good     | 2025-04-26 20:39:06 |
|          4 | 3        | good     | 2025-04-26 20:56:44 |
|          5 | 3        | amazing  | 2025-04-26 21:17:05 |
|          6 | 3        | superb   | 2025-04-26 21:35:47 |
|          7 | 3        | sabhkabs | 2025-04-26 23:23:23 |
+------------+----------+----------+---------------------+
7 rows in set (0.00 sec)

mysql> select * from StudentResumeCheckHistory;
+---------+----------+-----------------+-----------------------
```



```
mysql> select * from StudentFeedbackHistory;
+------------+----------+----------+---------------------+
| FeedbackID | Username | Feedback | Timestamp           |
+------------+----------+----------+---------------------+
|          1 | 6        | good     | 2025-04-26 23:17:38 |
|          2 | 6        | amazing  | 2025-04-26 23:21:42 |
+------------+----------+----------+---------------------+
2 rows in set (0.00 sec)
```

7.0.Test Procedure and Implementation

Here are example test cases categorized by module and functionality. Remember that these are just a starting point, and you'll need to create more detailed test cases based on your specific requirements and edge cases.

7.1.HR Module - Resume Ranking

Ranking Functionality Baseline:

Steps: Upload a job description and a few resumes (some highly relevant, some moderately relevant, some irrelevant). Trigger the ranking process.

Expected Result: Resumes ranked based on their relevance to the job description, with highly relevant resumes ranked at the top.

Support for Various Resume Formats:

Steps: Upload a job description and resumes in different formats (PDF, DOC, DOCX). Trigger ranking.

Expected Result: The system correctly parses and ranks resumes in all supported formats.

Keyword Matching:

Steps: Upload a job description with certain keywords and resumes that contain or do not contain these keywords.

Expected Result: Resumes with higher keyword matches applicable to the job description are ranked higher.

Semantic Matching:

Steps: Upload a job description and resumes that contain synonyms or related words for necessary skills and experience.

Expected Result: The system catches and takes semantic similarities into consideration, placing better resumes higher despite missing exact words.

Experience Matching:

Steps: Load a job announcement asking for special years of experience and resumes ranging from different experiences.

Expected Result: Resumes with experience numbers closer to required experience are listed higher.

Education Matching:

Steps: Load a job announcement asking for educational qualifications and resumes with distinct degrees and course studies.

Expected Result: Resumes with similar or closely related educational backgrounds rank higher.

Weights that can be customized:

Steps: Upload a job description and resumes. Set weights for skills, experience, and education. Start ranking.

Expected Result: The ranking order varies based on the weights used, giving more importance to the criteria with greater weights.

Filtering Ranked Results:

Steps: Rank resumes for a job. Use filters based on skills, level of experience, or education.

Expected Result: The list of resumes shown is filtered appropriately against the chosen criteria.

Empty Resumes/Job Descriptions:

Steps: Upload empty resume or empty job description. Start ranking.

Expected Result: The system responds to the empty input suitably, showing a suitable error message or no results.

Large Number of Resumes:

Steps: Upload large set of resumes (e.g., 100+). Start ranking.

Expected Result: The system analyzes and ranks the resumes in an acceptable amount of time (refer to your non-functional requirements).

7.2. HR Module - Soft Skill Ranking:

Basic Video Upload and Processing:

Steps: Upload an interview video. Start the soft skill analysis.

Expected Result: The system uploads and processes the video successfully.

Confidence Assessment (Visual Cues):

Steps: Upload videos where candidates show clear visual indicators of confidence (e.g., good posture, direct eye contact) and lack of confidence (e.g., slouching, fidgeting, avoiding eye contact).

Expected Result: Videos with positive confidence cues are assigned higher confidence scores.

Confidence Assessment (Vocal Cues):

Steps: Upload videos where candidates talk confidently and at a consistent rate (confident) and those who hesitate and pause frequently (less confident).

Expected Result: Videos with confident vocal cues are assigned higher confidence scores.

Tone Analysis (Positive/Negative):

Steps: Upload videos where candidates convey positive and negative emotions.

Expected Result: The system identifies and scores the tone correctly.

Tone Analysis (Professionalism):

Steps: Post videos with different degrees of professional tone.

Expected Result: Videos with professional tone have a better tone score.

Communication Clarity:

Steps: Post videos in which candidates give clear and brief answers compared to rambling or vague candidates.

Expected Result: Videos with clear answers have a better communication score.

Communication Structure:

Steps: Post videos in which candidates give well-organized answers compared to confused responses.

Expected Result: Videos with good communication structure are assigned a high communication score.

Weighted Soft Skills for Customization:

Steps: Upload a video. Change the confidence, tone, and communication weights. Start analysis.

Expected Result: Overall soft skill rank is according to the changed weights.

Support for Various Video Formats:

Steps: Upload videos in multiple supported formats (MP4, AVI, MOV).

Expected Result: The system handles videos in all supported formats.

Poor Audio/Video Quality:

Steps: Upload videos of poor visual or audio quality.

Expected Result: The system treats the poor quality gracefully, perhaps giving a warning or reduced confidence in the results.

7.3. Student Module - Resume Checker:

Basic Upload and Analysis:

Steps: Upload a resume. Trigger the checking process.

Expected Result: The system checks the resume and displays the results.

Grammar and Spelling Errors:

Steps: Upload a resume with grammatical and spelling errors.

Expected Result: The system detects and marks these errors, offering correct alternatives.

Formatting Issues:

Steps: Upload a resume with improper formatting (fonts, spacing, bullet points).

Expected Result: The system recognizes and recommends improvements for inconsistencies in formatting.

Content Analysis (Action Verbs):

Steps: Upload a resume with weak action verbs used in experience descriptions.

Expected Result: The system recommends stronger action words.

Content Analysis (Quantifiable Achievements):

Steps: Upload a resume with ambiguous descriptions of achievements.

Expected Result: The system recommends quantifying achievements using numbers and data.

Length and Conciseness:

Steps: Upload a resume that is excessively lengthy or has extraneous information.

Expected Result: The system recommends making the resume shorter and more specific.

Processing Different Resume Formats:

Steps: Upload resumes in many supported formats (PDF, DOC, DOCX).

Expected Result: The system successfully parses and analyzes resumes in all supported formats.

No Errors/Suggestions:

Steps: Upload a neatly formatted and well-written resume.

Expected Result: The system reports that no significant errors or suggestions were identified.

Large Resume:

Steps: Upload an extremely lengthy resume.

Expected Result: The system works through the resume in a reasonable time.

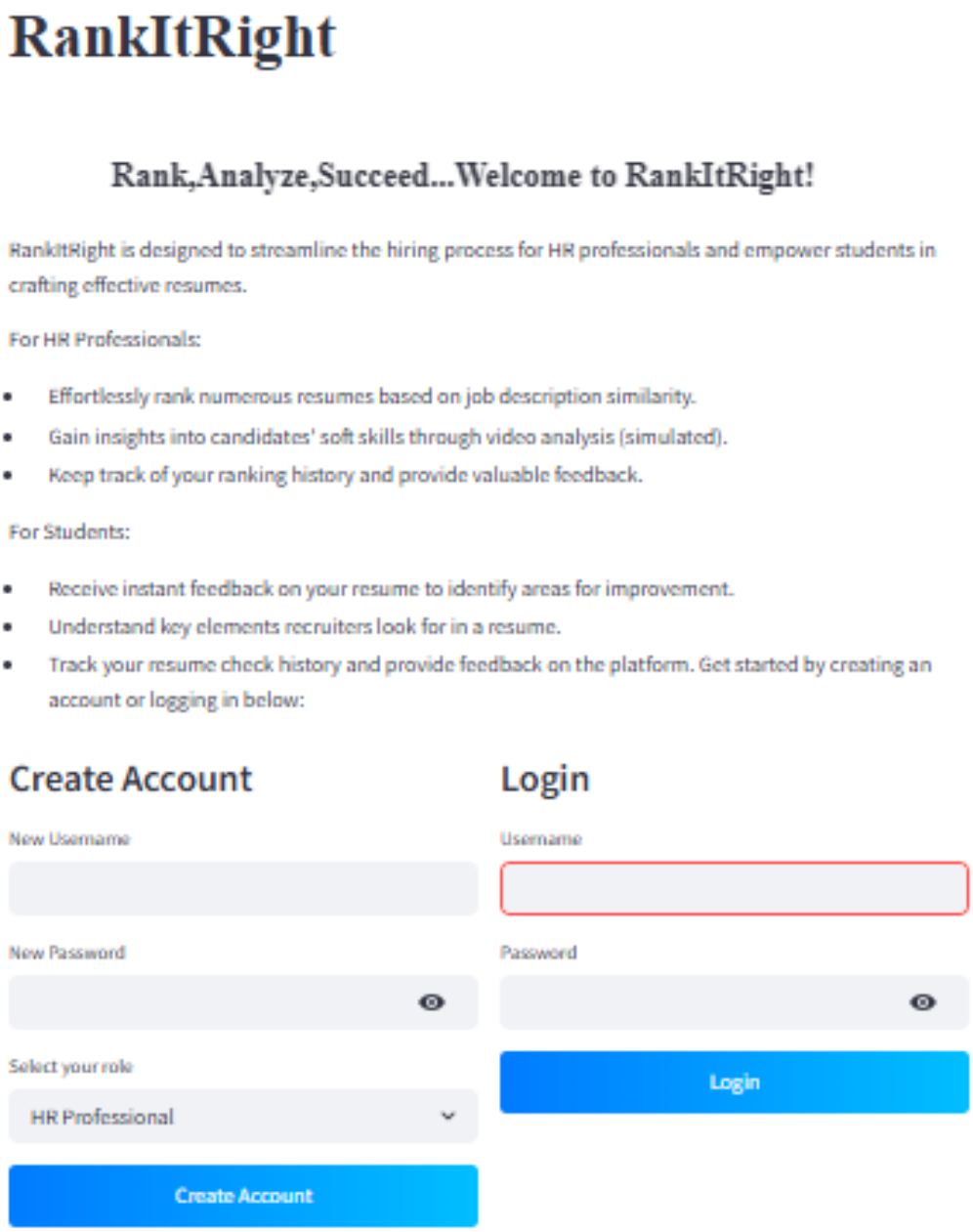
Empty Resume:

Steps: Submit an empty resume.

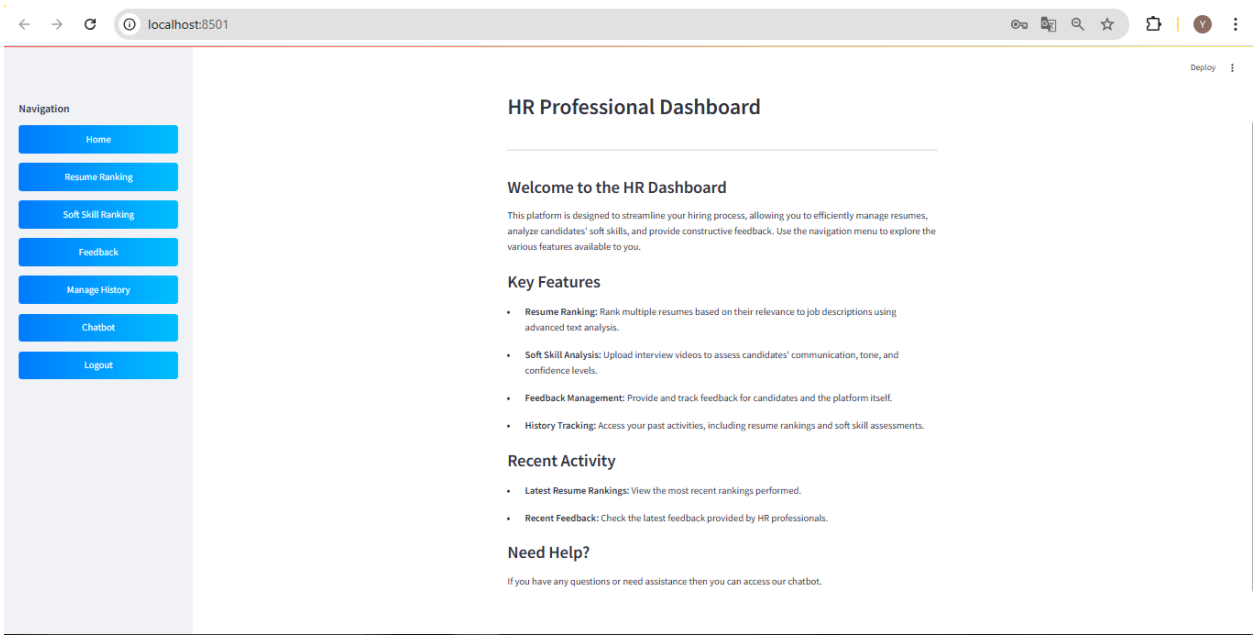
Expected Result: The system works through the blank input intelligently, reporting the correct message.

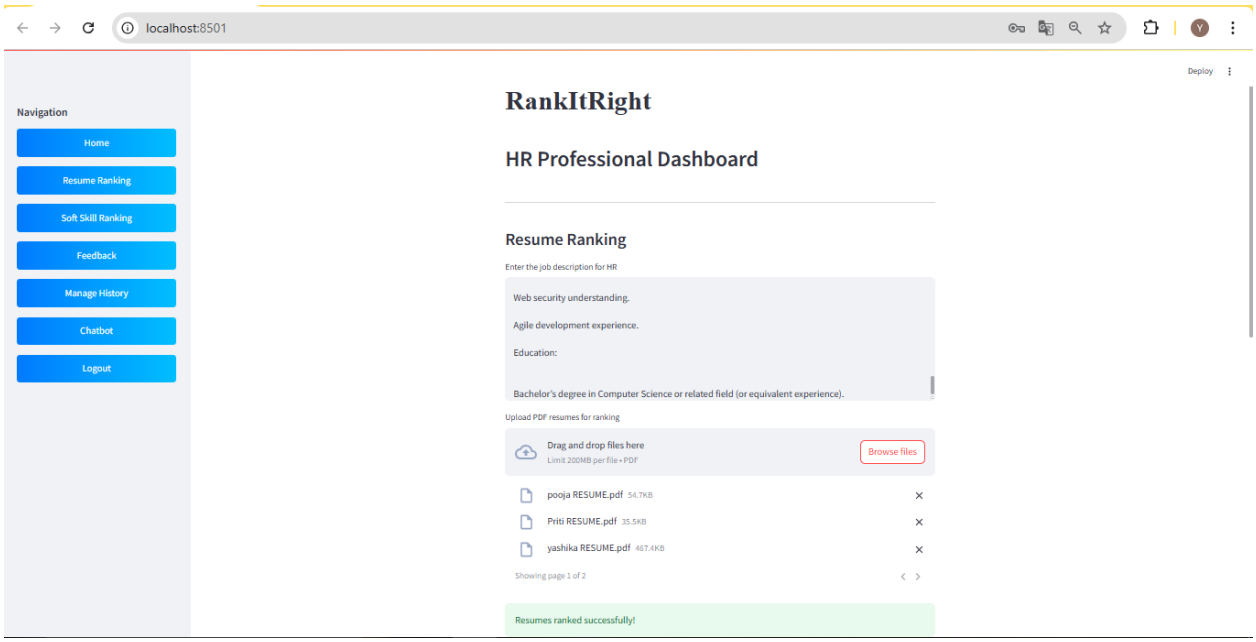
8.0.Screenshots

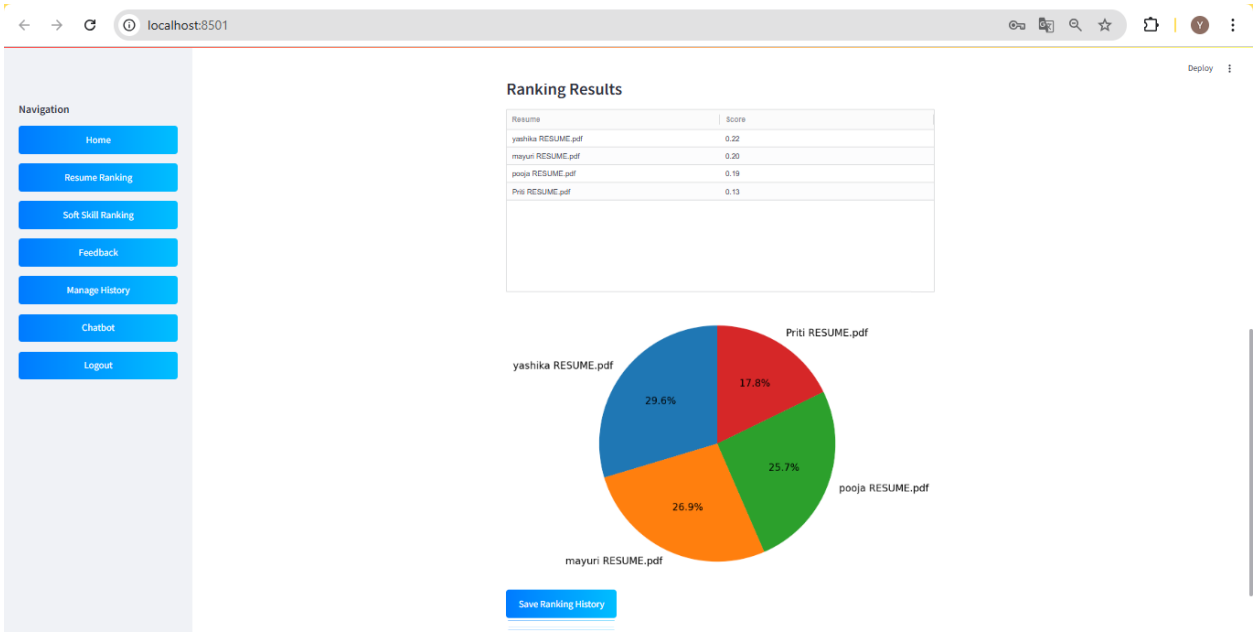
8.1 Main Page:

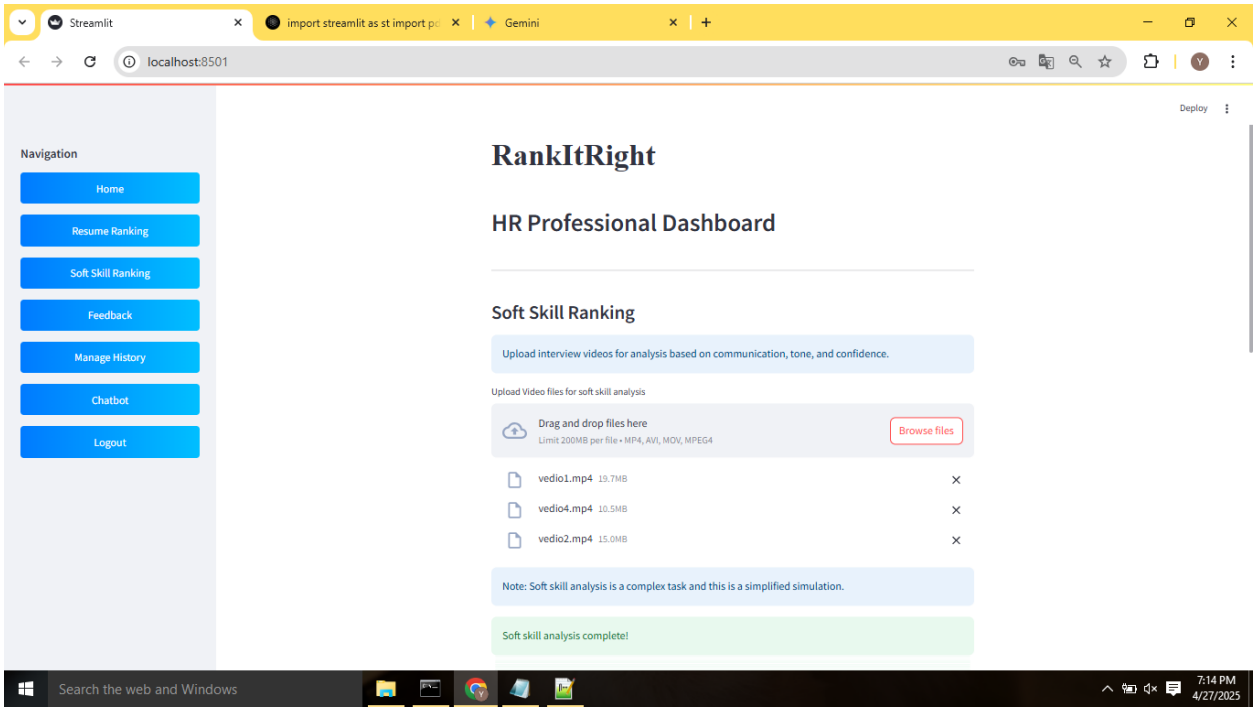


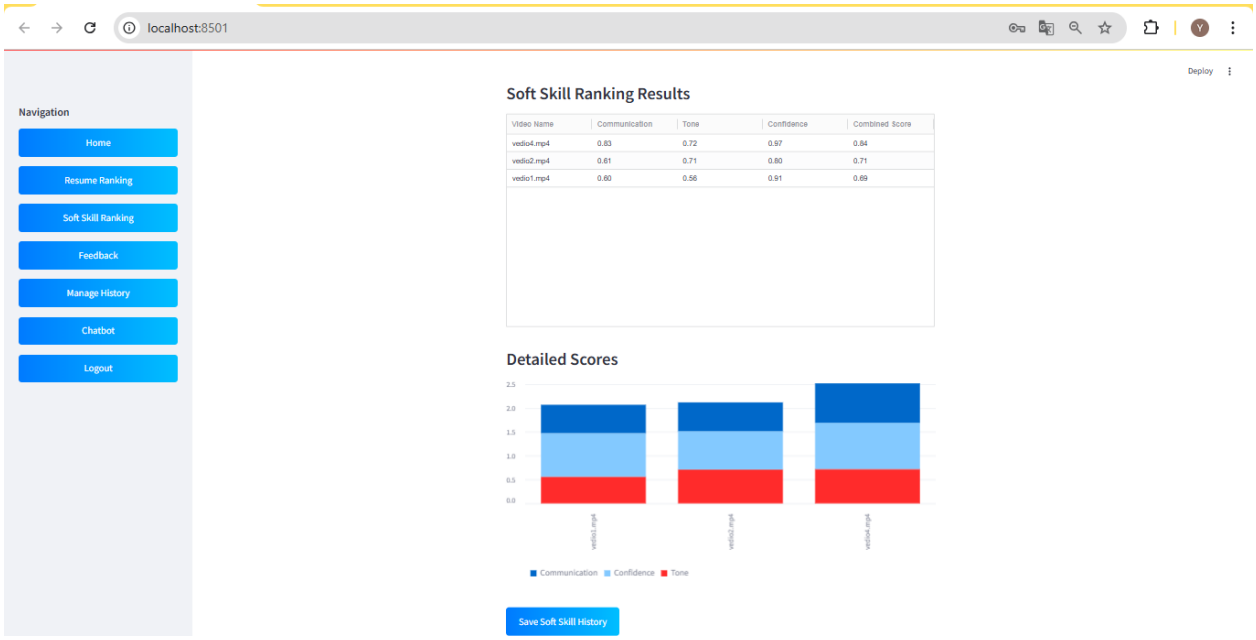
8.2 HR Dashboard:

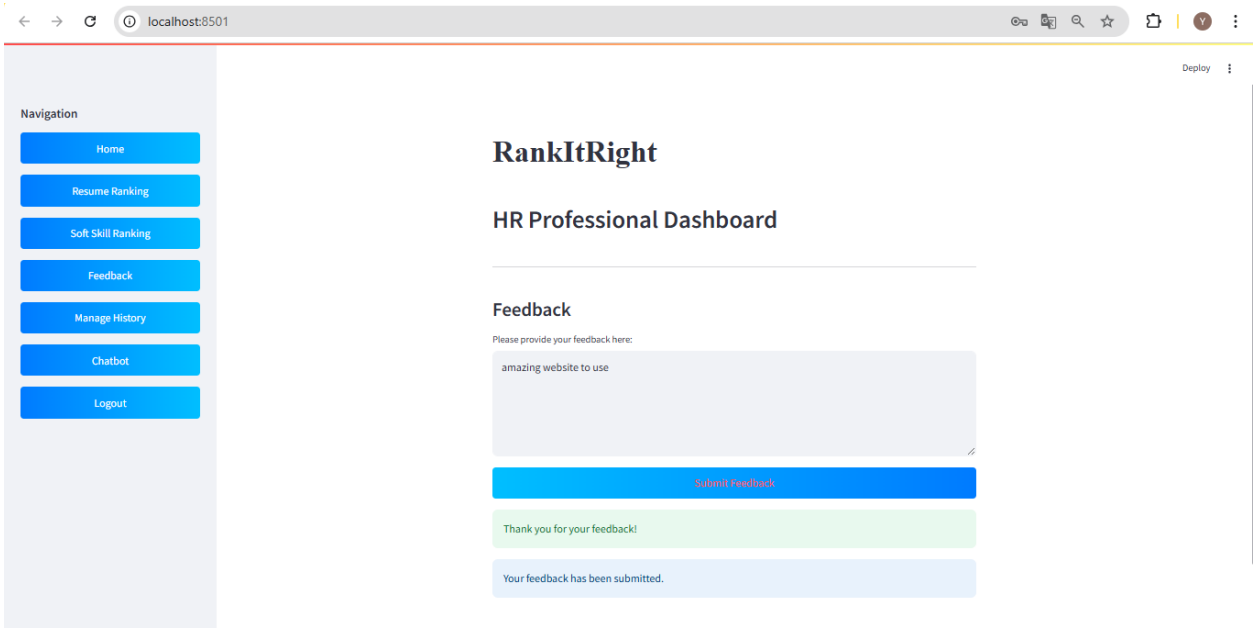


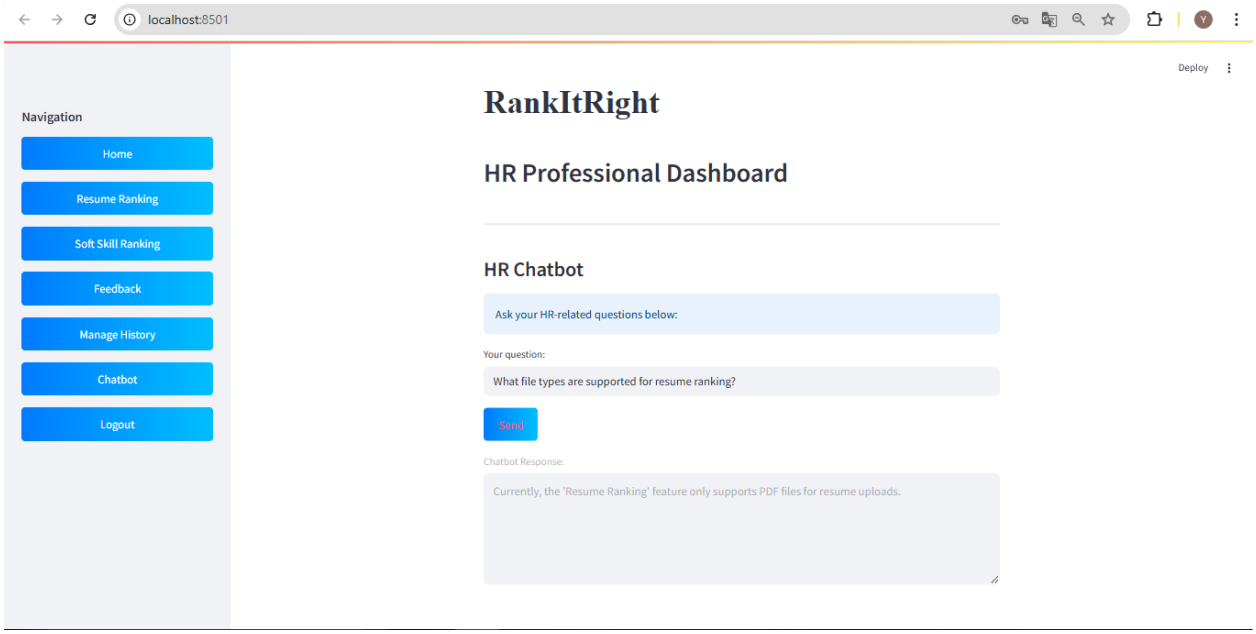



localhost:8501
Deploy
Navigation
Home
Resume Ranking
Soft Skill Ranking
Feedback
Manage History
Chatbot
Logout
Ranking Results
Resume
Score
yashika RESUME.pdf
0.22
mayuri RESUME.pdf
0.20
pooja RESUME.pdf
0.19
Priti RESUME.pdf
0.13
Priti RESUME.pdf
yashika RESUME.pdf
17.8%
29.6%
25.7%
pooja RESUME.pdf
26.9%
mayuri RESUME.pdf
Save Ranking History

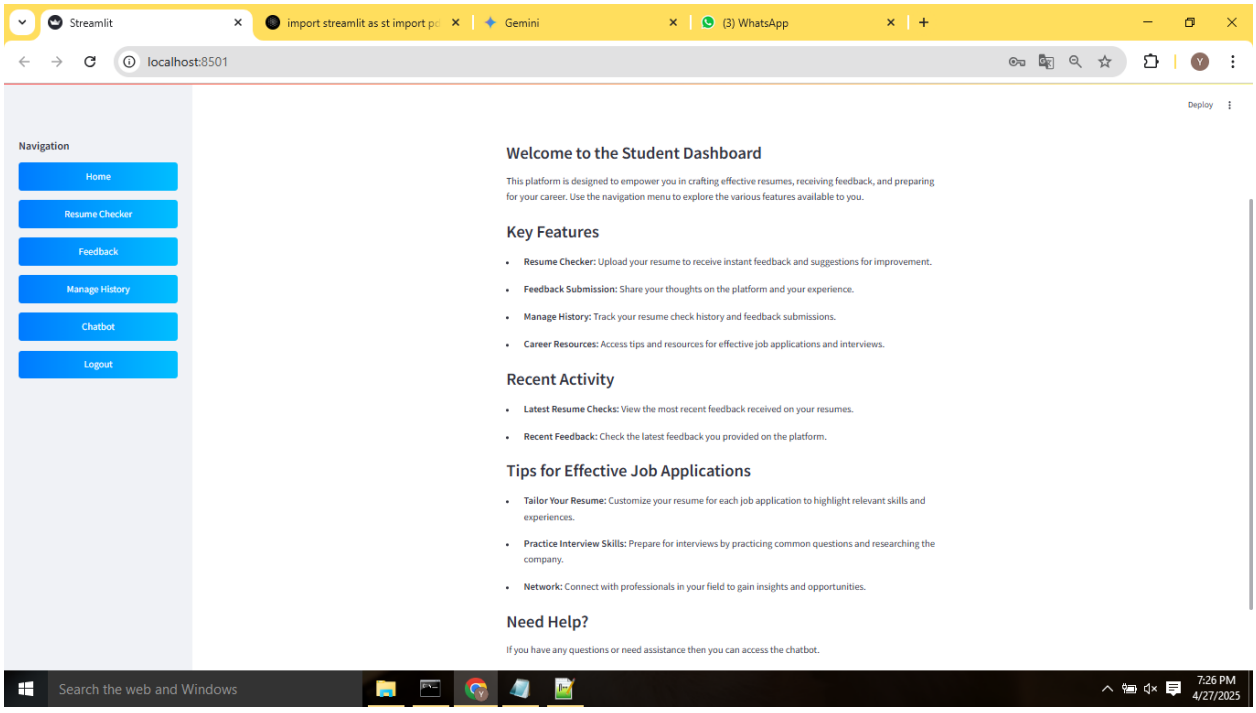

Streamlit
import streamlit as st import pd
Gemini
localhost:8501
Deploy
Navigation
Home
Resume Ranking
Soft Skill Ranking
Feedback
Manage History
Chatbot
Logout
RankItRight
HR Professional Dashboard
Soft Skill Ranking
Upload interview videos for analysis based on communication, tone, and confidence.
Upload Video files for soft skill analysis
Drag and drop files here
Limit 200MB per file • MP4, AVI, MOV, MPEG4
Browse files
vedio1.mp4 19.7MB
vedio4.mp4 10.5MB
vedio2.mp4 15.0MB
Note: Soft skill analysis is a complex task and this is a simplified simulation.
Soft skill analysis complete!
Search the web and Windows
7:14 PM
4/27/2025

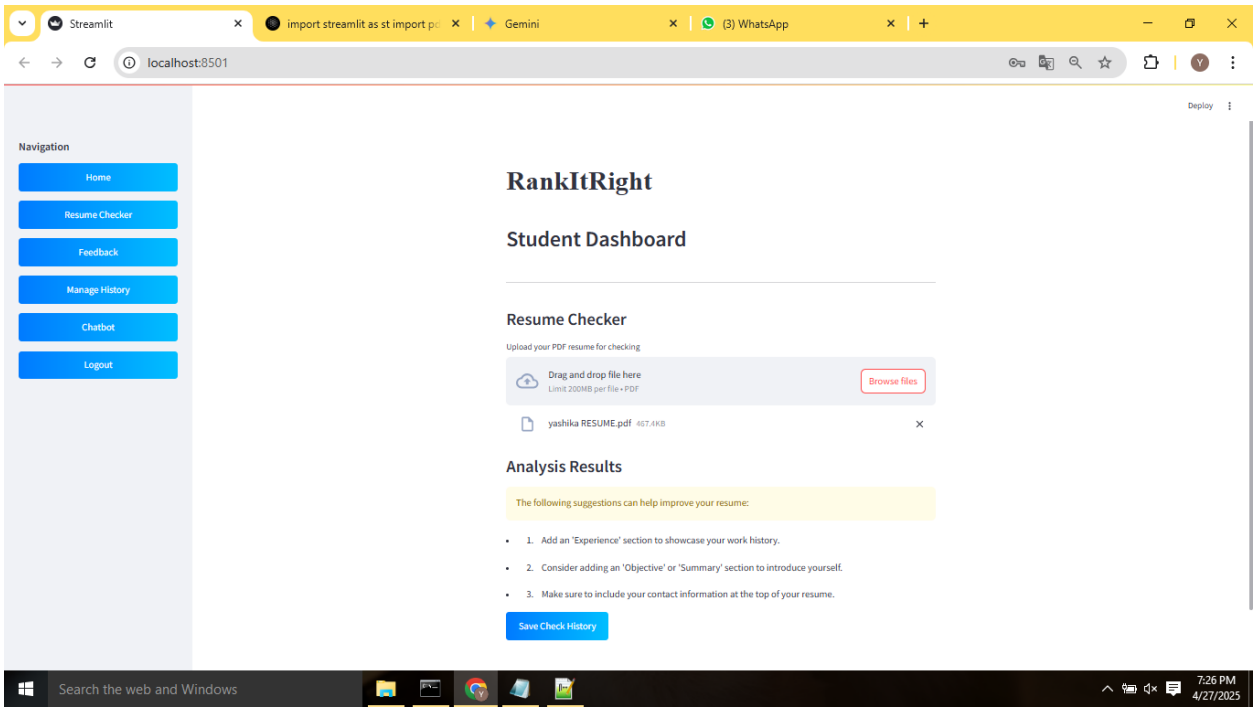

localhost:8501
Deploy
Navigation
Home
Resume Ranking
Soft Skill Ranking
Feedback
Manage History
Chatbot
Logout
Soft Skill Ranking Results
Video Name | Communication | Tone | Confidence | Combined Score
vedio4.mp4 | 0.83 | 0.72 | 0.97 | 0.84
vedio2.mp4 | 0.61 | 0.71 | 0.80 | 0.71
vedio1.mp4 | 0.60 | 0.56 | 0.91 | 0.69
Detailed Scores
2.5
2.0
1.5
1.0
0.5
0.0
vedio1.mp4
vedio2.mp4
vedio4.mp4
Communication
Confidence
Tone
Save Soft Skill History

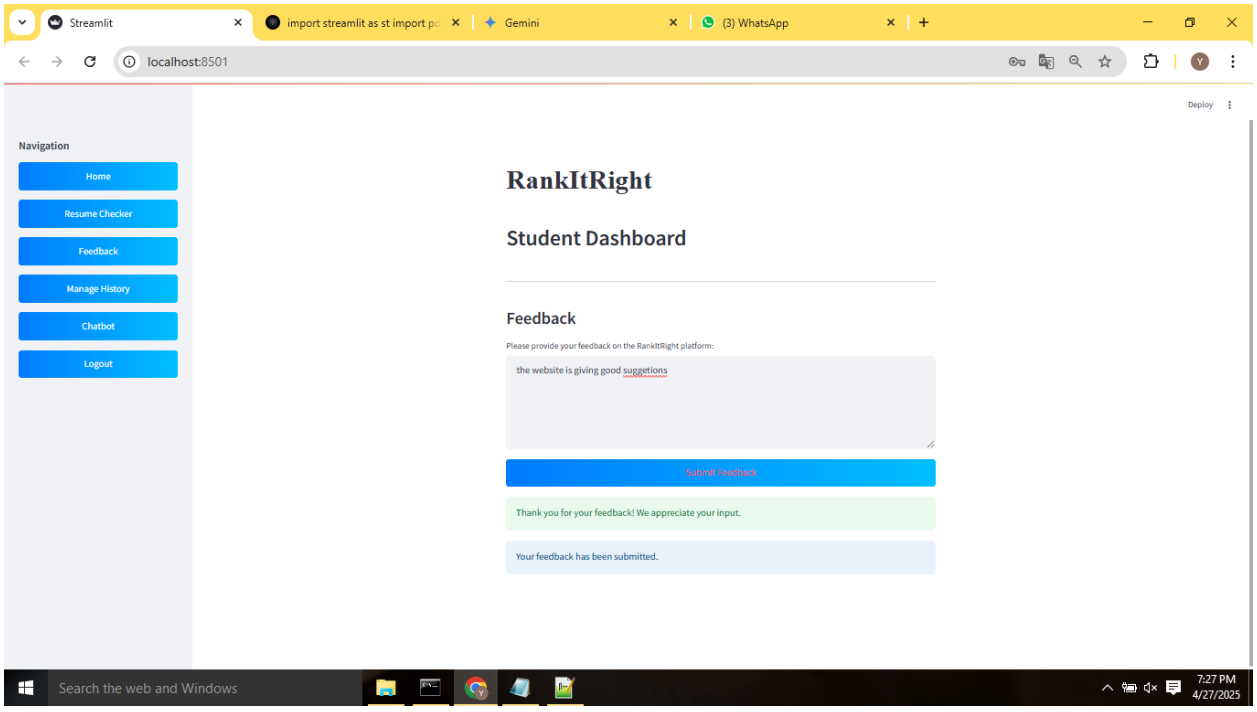

localhost:8501
Deploy
Navigation
Home
Resume Ranking
Soft Skill Ranking
Feedback
Manage History
Chatbot
Logout
RankItRight
HR Professional Dashboard
Feedback
Please provide your feedback here:
amazing website to use
Submit Feedback
Thank you for your feedback!
Your feedback has been submitted.

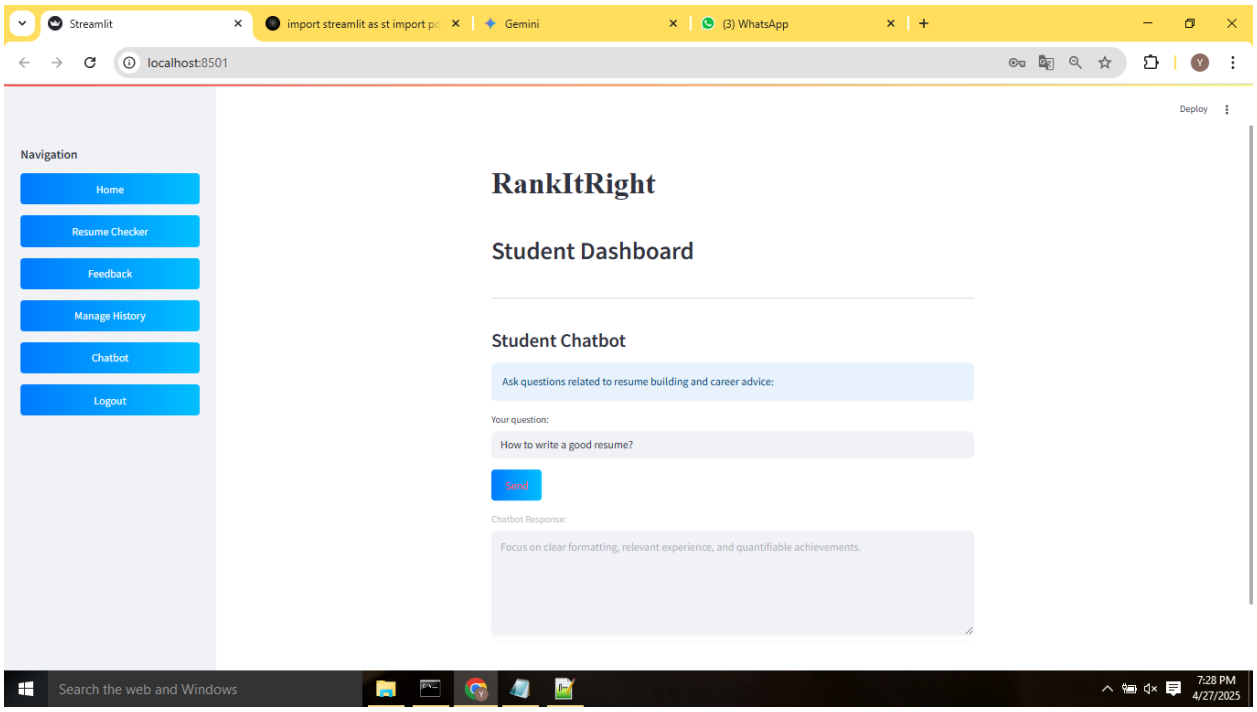


8.3 Student Dashboard




Streamlit
import streamlit as st import pd
Gemini
(3) WhatsApp
localhost:8501
Deploy
Navigation
Home
Resume Checker
Feedback
Manage History
Chatbot
Logout
RankItRight
Student Dashboard
Resume Checker
Upload your PDF resume for checking
Drag and drop file here
Limit 200MB per file • PDF
Browse files
yashika RESUME.pdf 467.4KB
Analysis Results
The following suggestions can help improve your resume:
1. Add an 'Experience' section to showcase your work history.
2. Consider adding an 'Objective' or 'Summary' section to introduce yourself.
3. Make sure to include your contact information at the top of your resume.
Save Check History
Search the web and Windows
7:26 PM
4/27/2025


Streamlit
import streamlit as st import pd
Gemini
(3) WhatsApp
localhost:8501
Deploy
Navigation
Home
Resume Checker
Feedback
Manage History
Chatbot
Logout
RankItRight
Student Dashboard
Feedback
Please provide your feedback on the RankItRight platform:
the website is giving good suggetions
Submit Feedback
Thank you for your feedback! We appreciate your input.
Your feedback has been submitted.
Search the web and Windows
7:27 PM
4/27/2025


Streamlit
import streamlit as st import pa
Gemini
(3) WhatsApp
localhost:8501
Deploy
Navigation
Home
Resume Checker
Feedback
Manage History
Chatbot
Logout
RankItRight
Student Dashboard
Student Chatbot
Ask questions related to resume building and career advice:
Your question:
How to write a good resume?
Send
Chatbot Response:
Focus on clear formatting, relevant experience, and quantifiable achievements.
Search the web and Windows
7:28 PM
4/27/2025

9.0.Advantages and Disadvantages

9.1.Advantages:

For HR Professionals:

- **Increased Efficiency in Candidate Screening:** Automating the initial stages of resume review and providing a ranked list significantly reduces the time and effort spent on manually sifting through a large volume of applications. This allows HR to focus on more strategic tasks and engage with the most promising candidates sooner.
- **More Objective and Consistent Analysis:** The system uses pre-defined standards and algorithms for resume ranking, which results in a more objective and consistent analysis process than entirely manual screening, which may be subjective and biased.
- **Holistic Candidate Assessment:** By combining soft skill evaluation from interview videos, the system gives a more holistic impression of the candidates apart from their paper qualifications. Assessing confidence, tone, and communication skills gives valuable insights into their potential as a fit for the job and the company culture.
- **Enhanced Quality of Hires:** By considering both hard and soft skills, the system identifies candidates who are not only qualified but also have the interpersonal qualities required to excel in the job and within the team. This can result in improved employee retention and overall team performance.
- **Customizable Assessment Criteria:** The capacity for HR users to modify the weights given to various resume criteria and soft skill dimensions enables the customization of the assessment process to suit the special needs of varying job positions and organizational requirements.
- **Data-Driven Insights into the Hiring Process:** The system can offer insights into how various keywords, qualifications, and soft skills contribute to successful hiring predictions, which enables HR to improve their job descriptions and interviewing process over time.
- Less Human Bias: Though the system is developed by humans and trained on data, it can be engineered to reduce unconscious biases that could happen during manual screening and subjective assessment.

For Students:

- **Enhanced Resume Quality:** The resume checker gives useful, automated feedback on content, grammar, spelling, and formatting, allowing students to make more polished, effective resumes that have a greater chance of getting noticed by recruiters.
- **Enhanced Employability:** By flagging and proposing corrections, the tool enables students to better highlight their qualifications and experience, thereby raising their chances of being shortlisted for interviews.
- **Understanding Recruiter Expectations:** The recommendations made are usually derived from general resume best practices and recruiter expectations, providing students with an understanding of what employers want.
- **Self-Paced Learning and Improvement:** Students can repeat the use of the tool as many times as necessary to iteratively refine their resumes at their own pace, creating a learning process that is ongoing.

- **Free and Accessible Resource:** Making this tool available to students adds value to the institution's value proposition and provides a useful free resource to aid their career advancement.
- **Emphasis on Key Resume Elements:** The tool directs students to emphasize key elements of a good resume, including the use of action verbs and quantifying accomplishments.

Overall Benefits:

- Taking Advantage of Python Technology: Using Python opens up a vast library ecosystem for NLP, machine learning, and multimedia processing, which makes it possible to create a high-level and dynamic system.
- **Dual-Sided Functionality:** Serving the requirements of both employers and job hunters makes a more complete and helpful system for the entire talent acquisition system.
- **Scalability Potential:** A properly designed Python-based system can be scaled to accommodate an increasing number of users and data.

9.2.Disadvantages:

For HR Professionals:

- strongly lies in the accuracy of the underlying algorithms and the quality of training data. Inaccurate or biased algorithms can cause skipping over qualified candidates or misreading soft **Dependence on Accuracy of Algorithms:** The success of the resume and soft skill ranking skills.
- **Potential for "Gaming" the System:** Candidates may learn how the system scores resumes and adapt their applications to include keywords and formats, even if their true qualifications or experience are not an exact match. Likewise, candidates may attempt to deliberately manipulate their behavior on video interviews to receive higher soft skill scores.
- **Over-Reliance on Automated Scoring:** There is a danger of HR professionals over-relying on the scores of the system and possibly missing out on nuances or intangible aspects that a human assessor would pick up. Human judgment is still essential in the ultimate hiring choices.
- **Data Privacy and Security Issues:** Processing sensitive information such as resumes and interview videos involves important data privacy and security issues. Strong safeguards must be in place to prevent unauthorized access or disclosure of this information, and applicable regulations must be complied with.
- **Technical Sophistication and Maintenance:** Building and sustaining the system involves specialized knowledge in NLP, machine learning, computer vision, and web development. Maintenance, algorithm updates, and debugging can be time-consuming and resource-intensive.

- **Initial Investment and Integration Costs:** Such a system may entail substantial initial investment in building, infrastructure, and integration with current HR systems.
- **Bias in Training Data:** If the training data used to build the ranking and assessment models reflect inherent biases (e.g., by gender, ethnicity, or socioeconomic status), the system could replicate or even magnify such biases in its assessments. Bias avoidance and mitigation through careful attention to data gathering is important.
- **Limited Contextual Understanding: Although** NLP methods are evolving, the system may continue to lack full understanding of the context and language subtleties of resumes and interview transcripts. Human assessors tend to pick up on subtleties that may elude an algorithm.

For Students:

- **Risk of Standardized Resumes:** Over-reliance on the suggestions of the resume checker may result in students making highly standardized, similar resumes that do not have personal touches and do not emphasize differential strengths.
- **Reliance on Computer-Generated Feedback:** Students may overly rely on the tool and not learn much about resume writing basics themselves.
- **Lack of Feedback Sensitivity:** Automatic feedback may lack sensitivity to the nuances of various industries or particular requirements of a job. Students may require additional human feedback to suit their resumes effectively.
- **Privacy Issues:** Personal resume data posted to the tool poses privacy issues, and students must have confidence that their information is managed responsibly and safely.
- **Narrow Scope of Feedback:** The student-facing module under development is limited to resume checking and does not offer feedback on soft skills or interview performance.

Overall Disadvantages:

Ethical Considerations: Automated systems in assessing individuals have ethical concerns regarding fairness, transparency, and accountability. Ensuring the system is responsibly and ethically used is important.
"Black Box" Problem: It is sometimes hard to interpret complex machine learning models, and thus it is hard to see why a given resume or interview got a certain score. This opacity can be problematic.

10.0.Conclusion and Future Enhancement:

10.1. Conclusion:

- This project endeavored to create a full-fledged Python-based system to solve major challenges in the talent acquisition process for both job seekers and HR professionals. Through the use of intelligent resume ranking and soft skill evaluation for HR, as well as a resume checker and suggestion system for students, we aimed to make the matching of talent with opportunity more efficient, objective, and ultimately more impactful.

- The module that interfaces with HR utilizes the strength of machine learning and natural language processing to automate the sometimes time-consuming process of resume screening and provides a more comprehensive candidate assessment by incorporating information on essential soft skills such as confidence, tone, and communication from interview videos. The weighting options provide HR professionals with the ability to customize the system to the requirements of various roles, making candidate shortlists more relevant and higher in quality.

- For students, the resume checker is a worthwhile, easy-to-use tool for self-enrichment. Through providing automated feedback on content, formatting, grammar, and spelling, the tool enables students to write more effective, more forceful resumes that more clearly present their qualifications and enhance their prospects of being interviewed.

- While recognizing the built-in difficulties and possible drawbacks of using automated evaluation systems, including the dependence on algorithm precision and the vulnerability to being "gamed," there are great benefits to be gained with this project. More efficiency for HR, more impartial and comprehensive candidate evaluation, and better resumes for students overall make for a more streamlined and fairer hiring environment.

- Future projects might involve continued improvement of the accuracy and stability of the algorithms, increasing the scope of soft skills measured, adding more sophisticated feedback for students, and resolving ethical issues related to bias and transparency. Ultimately, this project provides a basis for using technology to improve the talent acquisition process to the benefit of both hirers and seeker**s.**

10.2.Future Enhancement:

Building upon the current foundation, several future enhancements could significantly improve the functionality, scope, and impact of the Resume and Soft Skill Ranking System:

For the HR Module:

Advanced Soft Skill Analysis:

- **Non-Verbal Communication Deep Dive:** Use more advanced analysis of facial expressions (micro-expressions), body language (gestures, posture dynamics), and eye gaze patterns through advanced computer vision methods and possibly 3D modeling.
- **Emotion Recognition:** Incorporate emotion recognition to evaluate candidates' emotional intelligence and fit for jobs that involve certain interpersonal skills.
- **Behavioral Analysis:** Examine speech patterns and non-verbal behavior to make inferences about behavioral characteristics pertinent to job performance, including teamwork, leadership potential, and problem-solving style.
- **Cross-Cultural Communication Assessment:** Enhance the system's capacity to identify and factor in cultural differences in communication styles.
- **Integration with Applicant Tracking Systems (ATS):** Direct integration with major ATS systems would automate the HR process by enabling import of job descriptions and resumes directly and export of ranking and assessment results.
- **Real-time Interview Analysis:** Investigate the potential for giving real-time feedback or analysis during live video interviews (with proper consent), providing instant insight to the interviewer.
- **Personalized Interview Question Generation:** From the resume analysis and job description, the system might provide customized interview questions based on particular skills or areas of concern.
- **Predictive Analytics:** Process historical hiring data and performance metrics to determine patterns and enhance the accuracy of the ranking and soft skill assessment models over time, possibly forecasting candidate success in a particular role.
- **Feedback Loop for Algorithm Enhancement:** Have a mechanism for HR professionals to give feedback on the system's rankings and ratings, which can be utilized to retrain and enhance the underlying algorithms.
- **Multi-Language Capability:** Extend the system to support resume analysis and possibly soft skill rating in multiple languages.
- **Integration with Background Check Services:** Consider optional integration with background check services to offer a more complete candidate profile.

For the Student Module:

- **Soft Skill Evaluation (Practice Interviews):** Enable students to upload practice interview videos and get feedback on confidence, tone, and communication skills, like the HR module but for personal development.
- **Resume Template Library:** Provide a library of professionally prepared and industry-oriented resume templates that students can utilize as a reference point.
- **Job Description Alignment for Students:** Allow students to upload job descriptions they are considering and get an indication of how well their resume matches the qualifications, as well as recommendations on how to match their resume.
- **AI Resume Writing Assistant:** Add an AI chatbot or writing assistant that is able to take students through creating various parts of their resume, providing suggestions and examples in real-time.
- **Career Path Suggestions:** According to the skills and experiences indicated in the student's resume, the system may provide suggestions for potential career paths or related job functions.
- **Networking and Portfolio Management:** Enable students to connect their resumes with online professional profiles (e.g., LinkedIn) or portfolios.
- **Industry-Specific Resume Guidance:** Provide resume feedback and recommendations in accordance with the student's desired industry or job function.
- **Mentorship Link:** Potentially link students to alumni or professional mentors who may offer additional resume feedback and career guidance.

Overall System Improvements:

- **Enhanced User Interface (UI) and User Experience (UX):** Regularly improve the user interfaces for HR professionals and students to make them more usable and accessible.
- **Improved Security Measures:** Apply the most current security practices and protocols to guarantee the privacy and security of all user information.
- **Scalability and Performance Optimization:** Improve the system architecture to support an increasing number of users and data in an efficient manner.
- **API Development:** Create a fully featured API to enable seamless integration with other services and platforms.
- **Mobile Accessibility:** Make the HR and student modules accessible and easy to use on a mobile platform.
- **Accessibility Compliance:** Comply with accessibility requirements (e.g., WCAG) to ensure the system is accessible to people with disabilities.